

COMPUTER VISION | AT-2

Feature Matching for Sports Broadcast Camera Calibration

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Assessment	CO-3, AT-2

Input and Implementation



Reference: football overhead.jpg · 1920×1080



Broadcast: football broadcast.jpg · 2048×1365

INPUT

- Reference image: football overhead.jpg (1920×1080)
- Broadcast image: football broadcast.jpg (2048×1365)

IMPLEMENTATION

- SIFT detection & descriptor extraction
- FLANN matching + Lowe's ratio test
- RANSAC verification & homography
- Image warping and overlay

Input Images



SIFT



Feature Matching



Lowe's Ratio



RANSAC

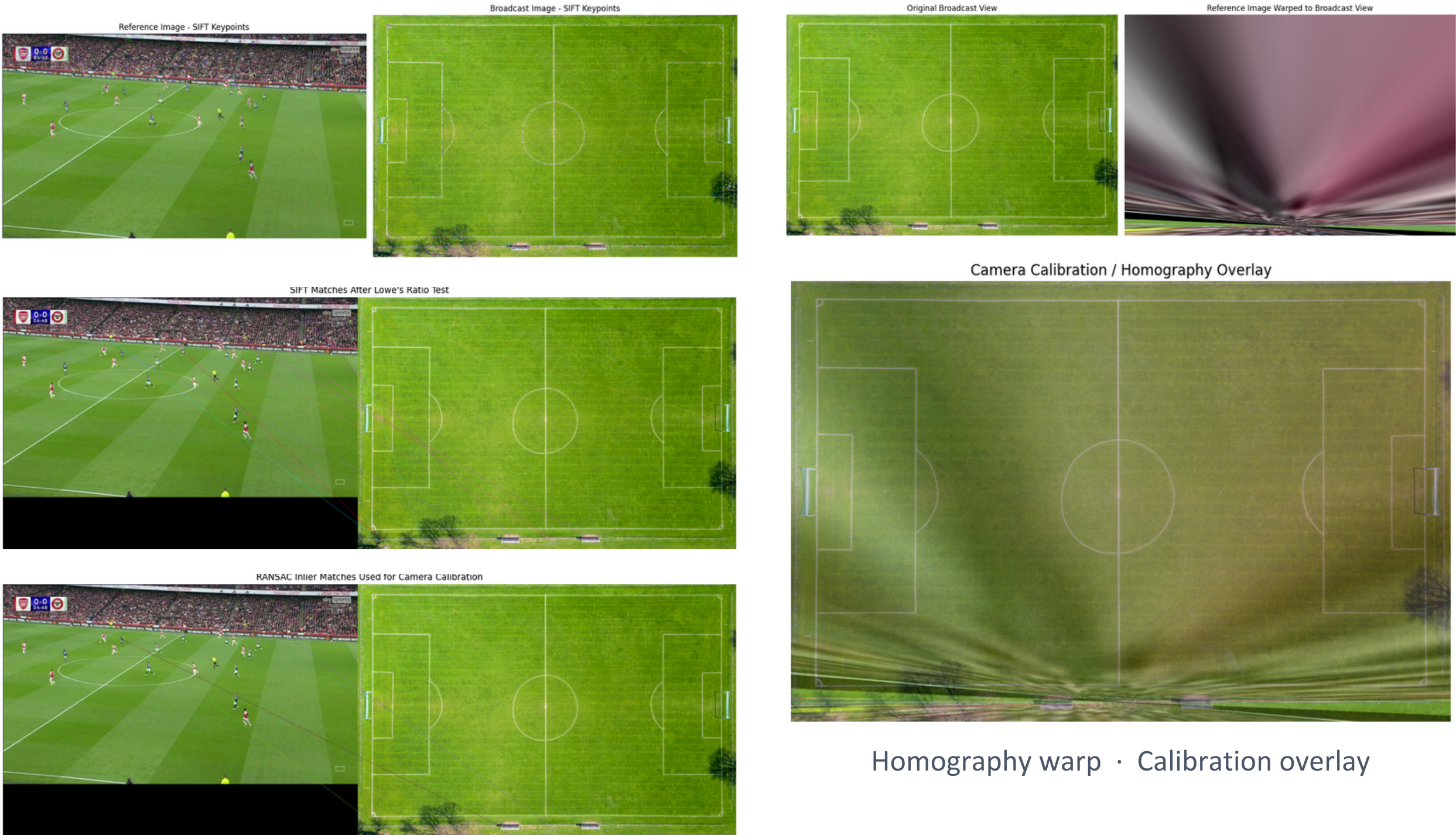


Homography



Calibration Output

Implementation Output and Results



Keypoints · Good matches · RANSAC inliers

Homography warp · Calibration overlay

RESULTS

Reference keypoints	3000
Broadcast keypoints	3000
Candidate matches	3000
Good matches (Lowe's)	10
RANSAC inliers	4
RANSAC outliers	6
Inlier ratio	40.00%
Mean reproj. error	0.0000 px
Median reproj. error	0.0000 px
Max reproj. error	0.0000 px

Observation and Conclusion

OBSERVATION

- 10 good matches obtained after Lowe's ratio test.
- RANSAC identified 4 inliers and 6 outliers.
- Inlier ratio: 40.00%.
- Reprojection error for inliers: 0.0000 pixels.
- The low inlier ratio indicates limited overlap or ambiguous/repetitive field features.

CONCLUSION

The implementation successfully demonstrated feature matching and geometric camera-view alignment between a reference sports-field image and a broadcast image.

FUTURE IMPROVEMENT

Use images with greater common field area.

Use more distinctive field features.

Test with multiple broadcast frames.